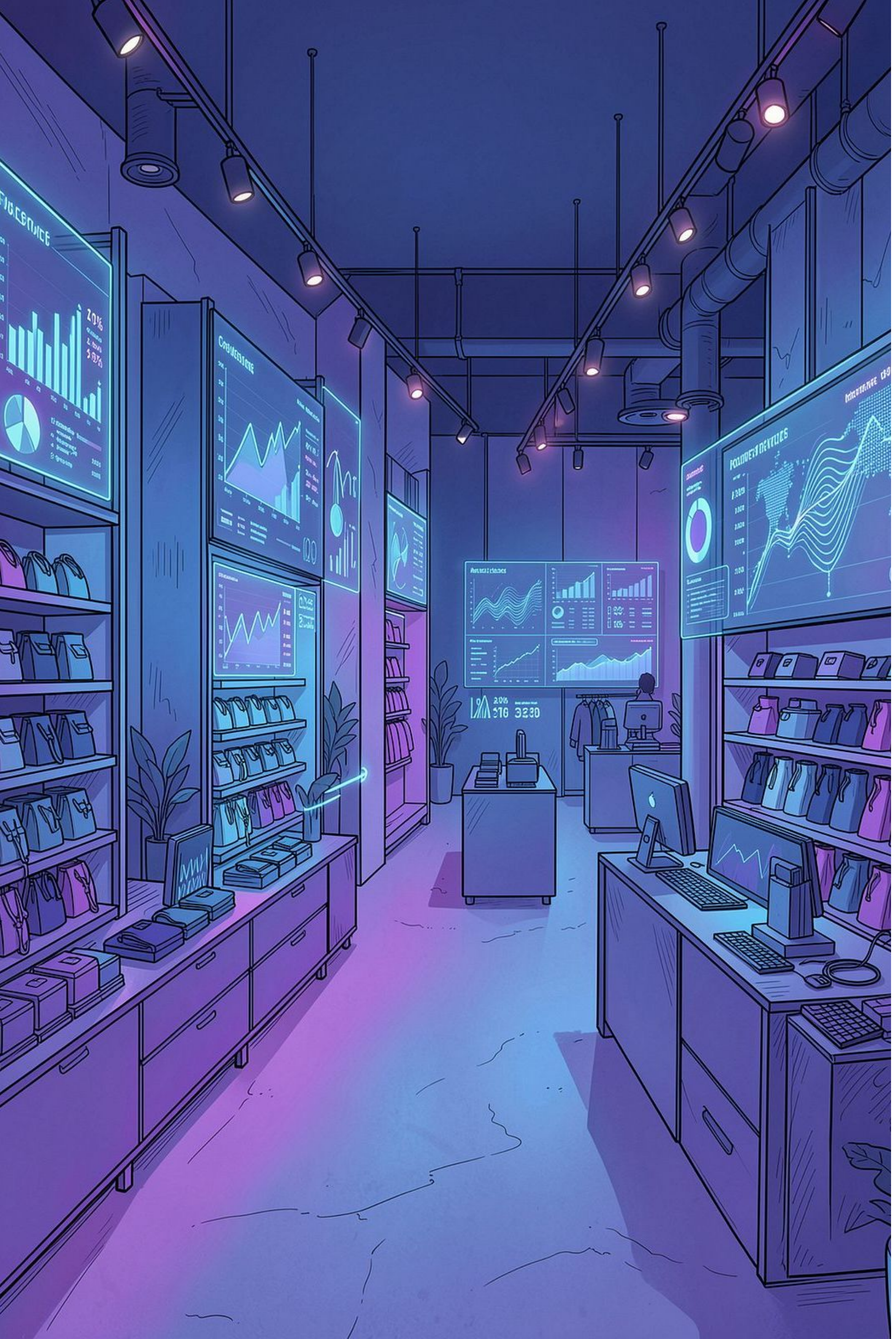


Retail Sales & Customer Analytics

A MySQL + Python + Power BI portfolio project analyzing **$2.30M in revenue**, **793 customers**, and **5,000 orders** to uncover where a retail superstore is losing money, which customers are at risk, and why.

PREPARED BY MOHAN

DATA ANALYST PORTFOLIO PROJECT

The Business Question

Superstore's leadership needed answers to three critical questions — this project was built to deliver them.

1	**Is revenue growing?** Track month-to-month trends and volatility across the full dataset.
2	**Which customers are at risk?** Identify the most valuable customers — and those most likely to churn.
3	**Which categories are least profitable?** Pinpoint underperforming product lines and understand the root causes.

Project Approach

A three-layer analytical pipeline — from raw data to interactive insights.

Phase 1

Database design and ETL with Python

Phase 3

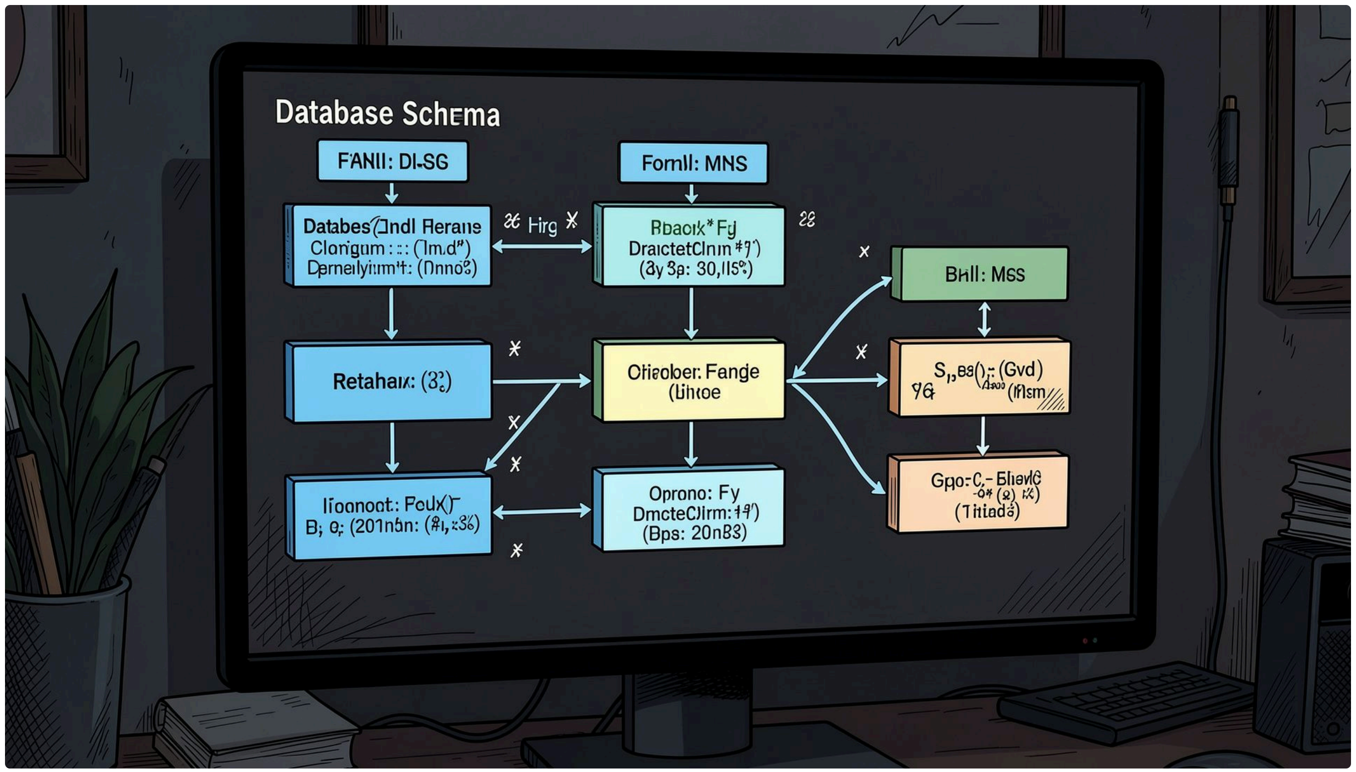
Power BI dashboard with ODBC and DAX

Phase 2

SQL analysis, RFM, cohorts and correlation

Each phase builds on the last: a clean relational schema enables rigorous SQL analysis, which feeds directly into a live-connected Power BI dashboard.

Database Design & ETL Pipeline



Normalized Relational Schema

Rather than a flat CSV, a four-table normalized schema was designed in MySQL:

- **Customers** — demographic and geographic attributes
- **Products** — category, sub-category, and pricing
- **Orders** — transaction headers with dates and regions
- **Order_Details** — line-item granularity linked via foreign keys

A Python ETL script handled staging, cleaning (mixed date formats, encoding issues), and loading — resolving Windows-specific MySQL driver challenges along the way.

SQL Analysis & RFM Segmentation

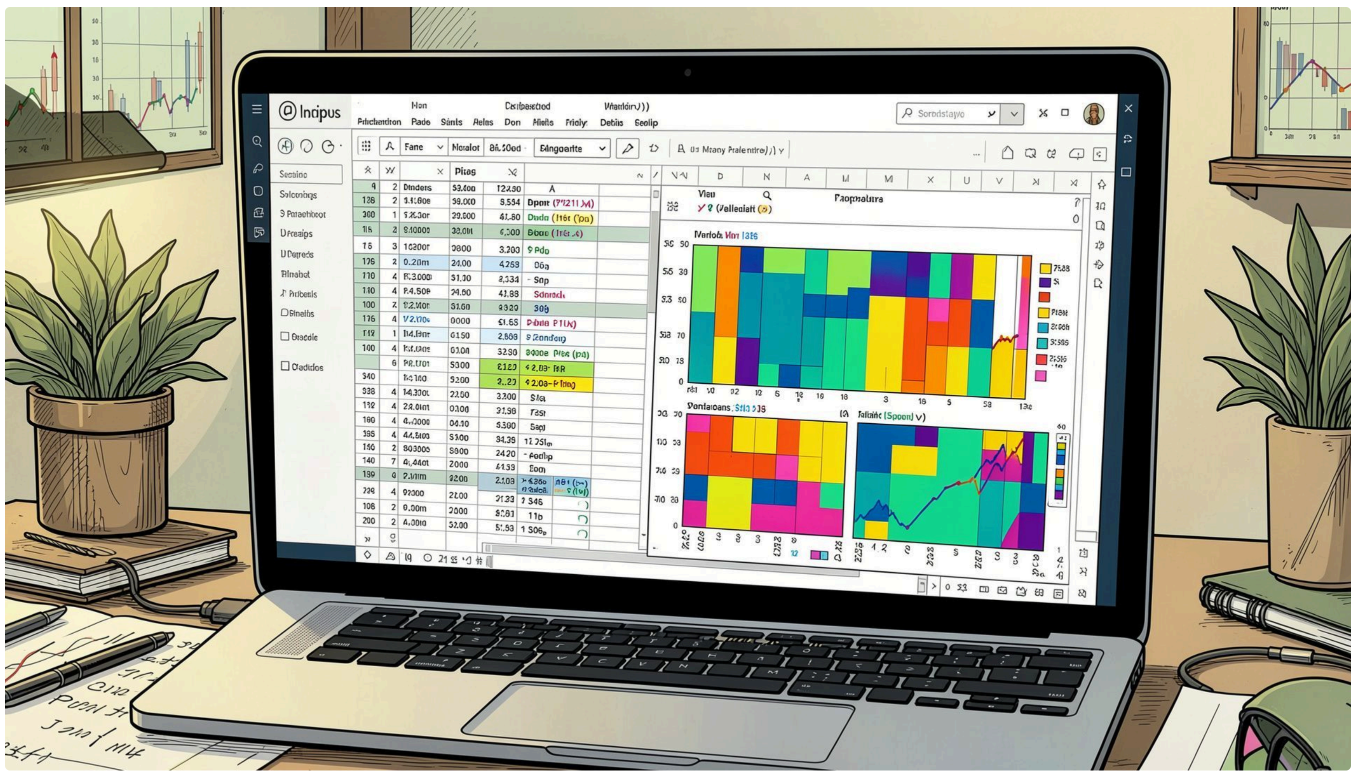
What the Queries Uncovered

SQL joins, GROUP BY, and window functions (NTILE) calculated monthly revenue trends, top products by profit, and category/region profit margins.

An RFM (Recency, Frequency, Monetary) model using CTEs classified all **793 customers** into five segments:

- **Best Customers** — 109
- **New Customers** — 179
- **Regular** — 364
- **At Risk** — 60
- **Lost / Churned** — 181

Python: Discount Correlation & Cohort Retention



Two Analyses Beyond SQL

Data was pulled via SQLAlchemy into Pandas for deeper exploration:

Discount vs. Profit Correlation

A correlation of **–0.22** confirmed that higher discounts erode margins — a relationship strongest within Furniture.

Cohort Retention Heatmap

Repeat purchase behavior tracked across **47 months** using Seaborn, revealing retention patterns by acquisition cohort.

Power BI Dashboard

Power BI connected directly to MySQL via ODBC. A relational data model mirrored the schema, and DAX measures — **Total Revenue**, **Profit Margin %**, and **Total Orders** — powered an interactive, cross-filterable single-page dashboard.

$2.30M

Total Revenue

$286K

Total Profit

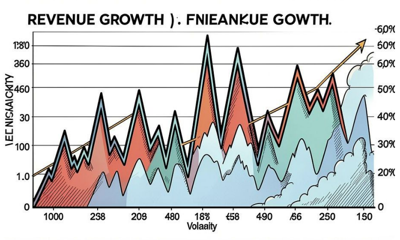
12.47%

Profit Margin

5K

Total Orders

Key Findings



Revenue Growing — But Volatile

Revenue grew steadily 2014–2017, but sharp month-to-month swings (e.g., $82K → $12K) suggest seasonality or inconsistent demand.

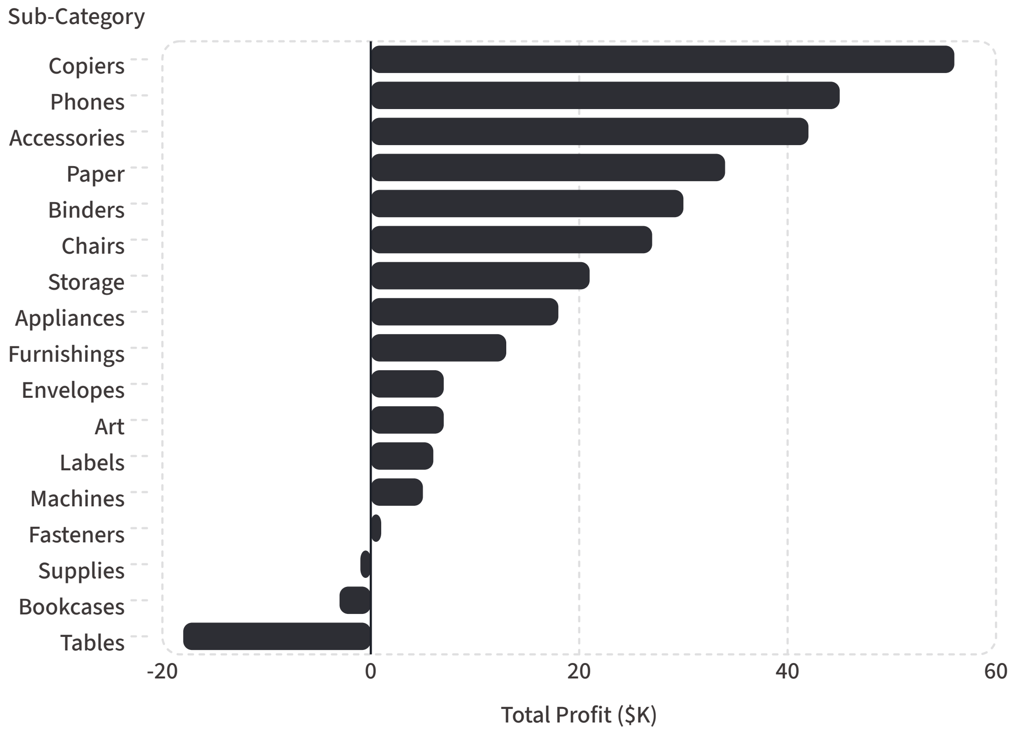
~30% of Customers Churned or At Risk

181 customers (23%) fully churned and **60 (8%)** at risk — nearly a third of the base, representing a major retention opportunity.

Furniture Weakest — Driven by Discounting

Furniture margins were negative in 2 of 4 regions (Central: –67%, East: –9%). Average discounts of **17.4%** — highest of any category.

Profit by Sub-Category



Furniture Sub-Categories Are Losing Money

While Technology and Office Supplies drive strong margins, Furniture sub-categories tell a different story. **Tables (–$18K)** and **Bookcases (–$3K)** are the only sub-categories with negative total profit — both dragged down by aggressive discounting that consistently exceeds the profitability threshold.

Tables and Bookcases alone account for **$21K in net losses** — the clearest target for pricing intervention.

Recommendations & Tools

Three Actions to Take Now

1 Cap Furniture Discounts Below ~30%

Especially for Tables and Bookcases, where profit turns consistently negative above this threshold.

2 Launch a Win-Back Campaign

Target the **241 customers** in At-Risk and Lost/Churned segments — prioritize At-Risk first given their proven purchase history.

3 Diagnose Revenue Volatility

Determine if monthly swings are seasonal (predictable) or signal inconsistent demand that marketing cadence could smooth.

Tools & Skills Demonstrated

MySQL

Schema design, joins, CTEs, window functions

Python

Pandas, SQLAlchemy, Seaborn — correlation & cohort analysis

Power BI

DAX measures, data modeling, ODBC connectivity, interactive dashboards